

Peer Influence on Decision-Making for Social Activity vs. Academic Performance

PSYCH 505: Discussion Section 306 - Group 3

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Introduction (*Written by: Kiera; Edited by: Isabella*)

The transition from adolescence to adulthood is a period of psychosocial changes that are heavily influenced by the environment, and students going off to college are often having their first unmonitored life experiences independent of caregiver supervision. During this time, decision-making and long-term goals are important for college students to keep in mind, and peer influence may play a role in affecting this process.

Kaleem et al. (2025) found that young adults tend to display low self-control and resistance to immediate gratification with peer influence, suggesting that peer influence could potentially moderate decision-making processes and reward value assignments, affecting one's ability to make decisions that support long-term goals.

Neurobiologically, we also know that the development of the prefrontal cortex, which is not fully formed during college-aged years, is vital to goal-directed decision-making and inhibition of impulses. In addition, a key area for reward processing, the ventral striatum, has higher activity in adolescents during receipt of reward with peers watching, suggesting that reward sensitivity is heightened at this developmental stage (Smith et al., 2015).

College-aged individuals are at the greatest predisposition for engaging in risky behaviors (Shulman & Cauffman, 2014), and social comparability is often heightened at this time. This adds to risky behavior dynamics and potentially drives shifts towards engagement in risky behaviors (Goethals & Zanna, 1979).

In this study, we aim to examine how peer influence will impact the prioritization of academics over “risky” behaviors such as going out with friends or not studying. Our research question asks, to what extent do behaviors of peers influence affinity for immediate social gratification versus long-term academic goals. By comparing peer influence through pre- and post- exposure conditions, we aim to expand our understanding of how college students are influenced by social pressure and the extent to which it affects their goal-directed decision-making.

Given that prior research has shown that peer influence can be a strong driver of risky decision-making (Patel & Bixter, 2025), risky behavior may increase with peer exposure. Based on previous findings, we expect that post-exposure to peer choice, a decrease in the prioritization of academics will occur when provided the opportunity to make a choice. Specifically, we hypothesize that the post-exposure to peer choice condition will increase the proportion of social choice selection compared to the pre-exposure condition.

Methods (*Written by: Isabella and Luke; Edited by: Kiera*)

Participants completed a web-based decision-making task that presented a series of hypothetical college-environment-based scenarios. Each scenario asked participants to choose between two options: one prioritizing an immediate social reward and the other prioritizing maintenance of good academic performance, benefiting a long-term goal. Social choices, which were considered the “risky” option in this task, included behaviors such as going out, skipping class, or avoiding studying. In addition, academic choices, which were considered the “non-risky” choice option, included behaviors such as studying, attending class, or preparing for an exam.

The task consisted of two phases: a pre-exposure phase and a post-exposure phase. In the pre-exposure phase, participants completed five scenarios without any information about peer behavior. These trials were designed to measure participants' baseline decision-making preferences in which choices were based only on their own individual goals or values. In the post-exposure phase, participants completed five additional scenarios that included peer-influence cues. These cues described the behavior of peers in the scenario, such as friends encouraging the participant to go out or information suggesting that many peers chose the social option (e.g. “85% of your classmates are at the bar”).

At the beginning of the task, participants were shown equivalent academic and social satisfaction scores; both scores were set to start at 50%. The scores changed after each decision based on the option that was selected. Choosing the academic option increased academic standing while decreasing social satisfaction by 10%,

whereas choosing the social option increased social satisfaction while decreasing academic standing. For example, if a participant chose to prioritize studying, their academic standing could increase from 50% to 60%, while their social satisfaction decreased from 50% to 40%. This scoring system was designed to make the tradeoff between immediate social reward and long-term academic performance visible throughout the task.

After completion, the participants' choices were coded numerically for analysis. Academic/non-risky choices were coded as 0, while social/risky choices were coded as 1. This allowed us to calculate the proportion of risky choices selected in each phase. For each participant, we calculated the average risky-choice rate during the pre-exposure phase and the average risky-choice rate during the post-exposure phase. Higher scores indicated a greater tendency to prioritize immediate social rewards over academic goals.

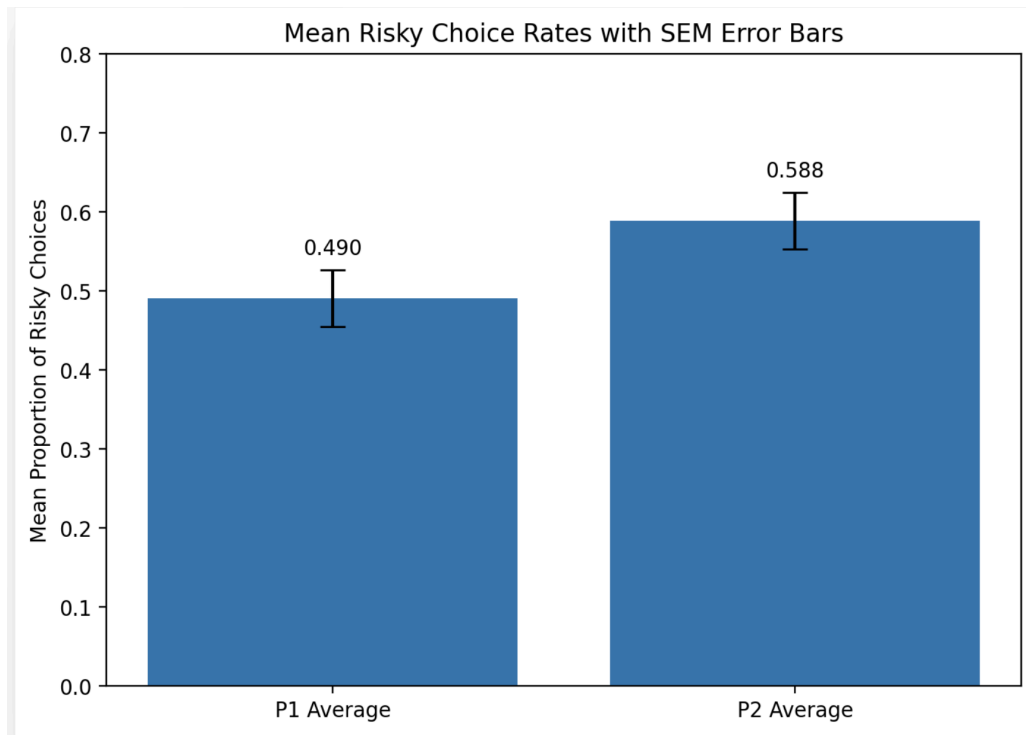
Data was analyzed using Google Sheets. We compared the average proportion of risky choices in the pre-exposure phase to the average proportion of risky choices in the post-exposure phase, finding mean and median values. To determine whether the observed change was statistically significant, we calculated a p-value comparing risky-choice rates across the two phases. The results were then visually graphed to display the change in social/risky choice selection from the first half of the task to the second half.

Results *(Written by: Luke; Edited by: Kiera and Isabella)*

Our results partially supported our hypothesis, as participants became more likely to select the social/risky option after peer exposure. To evaluate this shift, we compared risky choice rates before and after peer exposure, focusing on whether peer cues increased decisions such as going out, skipping class, or avoiding studying. Participants selected the social/risky option on an average of 49.0% of trials in the first half of the task. In the second half, the average increased to 58.8% of trials.

Figure 1

Mean Risky Choice versus Mean Proportion of Risky Choices



Note. This figure demonstrates the comparison of mean risky choice rates versus mean proportion of risky choices with standard error of the mean bars.

Participants chose the risky/social option more often in P2 ($M = 0.588$) than in P1 ($M = 0.490$), and this difference was statistically significant, $t(50) = 3.62$, $p < .001$, Cohen's $d_z = 0.51$, 95% CI [0.044, 0.152]. This indicates a medium-sized increase in risky/social choice selection after peer exposure.

The data suggest that peer exposure was associated with a reliable shift in decision-making within the task between P1 and P2. In context, this means students became more willing to choose options like socializing or going out even when those choices could come with academic costs. This supports the idea that peer cues can make immediate social rewards feel more appealing or socially acceptable.

Discussion (*Written by: Luke; Edited by: Isabella & Kiera*)

Our findings fit directly with prior research showing that peer influence can

increase risky decision-making, especially among college students. A study by Gardner & Steinberg (2005) found that peer presence increased risk-taking, risk preference, and risky decision-making, with the effect being especially relevant for younger participants. This connects closely to our experiment, as our participants were college students making decisions that involved a tradeoff between immediate social rewards and longer-term consequences.

Although our effects were modest, the direction of the results fits into the broader idea that social context can shift people toward riskier choices (Gardner & Steinberg, 2005). Our findings also fit with research showing that peer influence can increase risky choices even when people understand the potential negative consequences. This is important for our study because participants knew the academic tradeoffs, but peer exposure may have still made social options feel more appealing (Smith et al., 2014).

From a neuroscience point of view, the increase in social and risky choices can be viewed through the lens of reward sensitivity and value-based decision-making. Although our study did not directly measure brain activity, prior neuroscience research suggests that peers can increase risk-taking by making risky choices feel more rewarding. Chein et al. (2011) found that peer presence increased adolescent risk-taking by heightening activity in reward-related brain regions, including the ventral striatum and orbitofrontal cortex.

In our study, peer exposure may have increased the subjective value of the social option, making immediate social rewards feel more salient than delayed academic consequences (Chein et al., 2011). To add, participants in our study were repeatedly choosing between an immediate social reward and a longer-term academic benefit. The peer cue may have shifted the reward value of the social option. Rather than showing that participants ignored academic consequences, the results suggest that social information may have changed how those consequences were weighed against immediate rewards.

Our results also connect to impulsivity since the risky choices in the task involved immediate benefits and delayed costs. The social options, like going out or joining

friends, provide quick social rewards, while the academic considerations are delayed and less emotionally immediate. Furthermore, Kaleem et al. found that peer pressure is related to risky decision-making among young adults and that impulsivity may help explain this relationship. Our study demonstrates that peer exposure potentially increases the pull of immediate gratification, making short-term social rewards available (Kaleem et al., 2025).

Limitations (*Written by: Julia & Luke; Edited by: Kiera*)

Though our study found that participants chose the risky/social option more often in P2 than in P1, a statistically significant difference ($p = < .001$), the effect in our study was modest. One possible reason for not having complete confidence in the results is the small sample size, since the study included only 51 participants. With a larger sample, the study may have had more statistical power to detect a reliable effect.

This study utilized a survey-based web design, where participants responded to hypothetical scenarios rather than real-life scenarios. Because of this, the peer influence they experienced was artificial and may not fully capture how people actually respond in real social situations. This could have affected the strength or accuracy of the results.

This study did not include an opposing peer-influence condition. Participants were not presented with scenarios where peers encouraged academic behaviors, such as choosing to study. Since peer influence was only represented in a socially rewarding direction, this limits our ability to compare how different types of peer pressure might shape decision-making. Because of this, the peer cue may have been weaker than peer pressure in a real college setting.

There was also a potential issue in the visual design of the task. Social choices were displayed with green buttons, while academic choices were shown in red, which could have influenced responses due to common color associations. While this may have been an implication, it was slightly intentional to simplify the task and help account for the study's relatively small sample size. Further analysis of color choice in terms of it being a mediating variable would give us greater insight to the effect of this.

There is also a lack of demographic data, including demographic variables such as age, year in school, gender, and additional background information. Inclusion of this data would allow for a more nuanced understanding of how decision-making processes may vary across groups. Specifically, gender differences and cultural norms would likely have an influence on responses to peer influence and reward sensitivity. Future research could address this by using a cross-sectional design to compare participants from diverse cultural backgrounds, helping to identify whether these effects are consistent or vary meaningfully across populations.

Future Directions (Written by: Julia & Kiera; Edited by: Kiera)

Future studies could build on previous research and our findings by incorporating neuroimaging techniques, such as fMRI, to further examine the neural mechanisms underlying decision-making. In particular, investigation of ventral striatum activity during risk-taking tasks could be examined to determine how risk predictions and reward value encoding are related. The study by Smith et al. (2015) highlighted how striatal engagement suggests that risk-taking may rely on calculations of risk predictions and prediction errors rather than perceived value of engagement in risky behavior. Further exploration of this decision-making pathway, utilizing brain imaging techniques would provide further insight to neural mechanisms. In addition, more examination of prefrontal cortex activity would be beneficial, specifically utilizing comparison of activation between college-aged individuals and older adults.

Another pathway of exploration should include the examination of decision-making frameworks, which could provide further insight to what leads to risky behaviors. Wolff & Crockett (2019) emphasize how research in risky decision making tends to define the decision as the behavior itself rather than separating the decision process from the selected behavior. This limits the ability to examine dual processes in decision making, leaving a gap in different decision making pathways utilized in real life.

Overall our findings and previous research demonstrates that peer presence during can modulate reward-related neural activity beyond risk-taking contexts, with important implications for both prosocial and risk-related behaviors (Smith et al., 2014;

Chein, et al. 2011). Using these foundations for further examination, future research should extend this work by focusing more specifically on college-aged participants (18–23), utilizing brain activity comparison to older adults, dual processing frameworks, and examination of differences in gender and background. These approaches will help clarify developmental differences in frontostriatal circuitry and provide a deeper understanding of the neural processes underlying risky decision-making in college populations and the moderating role of peer influence.

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